

Havells ERP+CRM

Enterprise Resource Planning & Customer Relationship Management System

Streamlining Business Operations, Inventory, and Vendor Management

Problem Statement

- **Fragmented Data:** Tracking sales, inventory, and vendor POs across disparate systems leads to inefficiency.
- **Manual Data Entry:** Extracting bill info manually is time-consuming and prone to human error.
- **Limited Visibility:** Lack of a unified dashboard for admin and employees hampers quick decision-making.
- **Complex Operations:** Managing societies, contractors, leads, quotations, and ticketing is chaotic without a unified workflow.

Tech Stack

Frontend: Laravel Blade Templates, TailwindCSS v4 with Vite, Axios

Backend: PHP 8.2 & Laravel 12, Spatie Permissions & ActivityLog, DomPDF

Database & Hosting: MySQL, Database Queue/Session

Architecture / System Design

- **Frontend Interface:** Users interact with the Laravel Web App built with Blade and Tailwind.
- **Business Logic:** Laravel Controllers handle the core business logic.
- **Database:** Data is stored and retrieved using Eloquent ORM interacting with a MySQL Database.
- **External Services:** Includes OCR Parsing Service for GST Bills and DomPDF for generating invoices and quotations.

Key Features

- **Comprehensive Dashboard:** Admin & Employee portals with Role-Based Access Control (RBAC).
- **Vendor PO & Invoice Management:** Create, track, and manage vendor purchase orders and invoices.
- **Smart OCR Integration:** Auto-parse GST bills (photos) to auto-fill PO details.
- **Inventory & Warehouse Management:** Stock reconciliation, stock-in/out, and transactions logging.
- **CRM & Ticketing:** Manage leads, customers, appointments, and support tickets.
- **Accounting:** Journal entries, balance sheet, and profit-loss reports.

Code Highlights

OCR Bill Parsing Logic (VendorPoController.php):

```
public function ocrParseBill(Request $request) {  
    // Validates uploaded image/PDF  
    // Mocked extraction simulating an OCR integration for GST invoices  
    $extractedData = [  
        'vendor_name' => 'Proxima Piping Systems',  
        'subtotal' => 35.71,  
        'grand_total' => 42.00,  
        // ... Parses item details, GST percent, HSN codes  
    ];  
    // Auto-creates supplier if not found  
    $vendor = Supplier::firstOrCreate(['name' => $extractedData['vendor_name']]);  
    return response()->json(['success' => true, 'data' => $extractedData]);  
}
```

Automatically populates complex PO forms from an uploaded bill.

Database Schema Overview

- **Users & Auth:** users, roles, permissions, activity_log
- **CRM Core:** customers, leads, societies, appointments
- **Inventory:** products, warehouses, stock_transactions, material_logs
- **Sales & Operations:** quotations, invoices, tasks, tickets, shipments
- **Accounts:** accounts, journal_entries, payments, credit_debit_notes
- **Vendors:** contractors, suppliers, vendor_pos, vendor_invoices

Challenges & Solutions

- **Challenge:** Creating complex POs from supplier bills is highly error-prone.

Solution: Implemented OCR endpoint that auto-reads GST bills and maps items, HSN codes, and taxes to the PO creation form.

- **Challenge:** Managing multi-tenant/branch data and diverse user roles (Admin vs. Employee).

Solution: Integrated spatie/laravel-permission, assigned branch-specific filters, and implemented strict middleware routing.

- **Challenge:** Complex tax and accounting reconciliation.

Solution: Built custom journal entry controllers and double-entry accounting tables linked directly to invoices.

Future Scope

- **Real OCR Integration:** Connect the mocked OCR endpoint to AWS Textract or Google Cloud Vision API for dynamic real-world invoice reading.
- **AI-Powered Analytics:** Use LLMs to generate insights from sales trends and customer tickets.
- **Mobile Application:** A native React Native / Flutter app for field agents (contractors/employees) to manage appointments and upload bills on the go.
- **Automated Tax Filing:** Direct API integration with government GST portals.

Conclusion

- **Havells ERP+CRM** is a robust, end-to-end platform tailored for scalable enterprise operations.
- By combining **inventory tracking, automated accounting, and OCR-assisted data entry**, it significantly reduces operational overhead.
- Built on modern **Laravel 12** and **TailwindCSS**, it is fast, secure, and ready for future integrations.

Thank You!